

Cyber Physical System Security

SC4015/CE/CZ4055

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CCDS, NTU



Contents

Course Organization

- Cyber Physical Systems
- Security
- Discussion



Tutor and Assistants

- **Lecturers**

- Anupam Chattopadhyay (anupam at ntu.edu.sg)

- **Teaching Assistants**

- Sayan Das (sayan005 at e dot ntu dot edu dot sg)
- Prasanna Ravi Kant (prasanna dot ravi at ntu dot edu dot sg)
- Peizhou Gan (peizhou dot gan at ntu dot edu dot sg)
- Kamal Raj (kamal dot raj at ntu dot edu dot sg)



Schedule

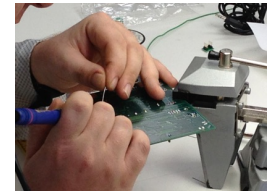
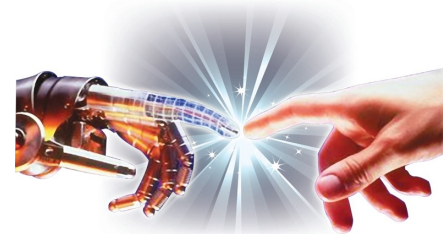
- **Lectures**
 - Thursdays 14:30 - 16:20 PM (LT1A)
- **Tutorials** (*starts from Week 3*)
 - Thursdays 16:30 - 17:20 PM (LT1A)
- **Laboratory Assignments** (*starts from Week 8*)
 - Tuesdays 12:30 – 16:20 PM (TEL1, TEL3 – HWLAB1)
 - Fridays 12:30 – 16:20 PM (TEL2, TEL4 – HWLAB1)
- (*check for updated schedule in NTULearn*)

What will we learn?

- Cyber Physical Systems are everywhere around us
 - You are using it
 - You will possibly *design/secure* it as part of your future career
 - You may end up discovering/innovating/using it



- We will learn
 - **Basic principles** of their design
 - How to address the **Security issues**
 - **Abstract concepts** with tutorials
 - **Practical concepts** with prototyping boards

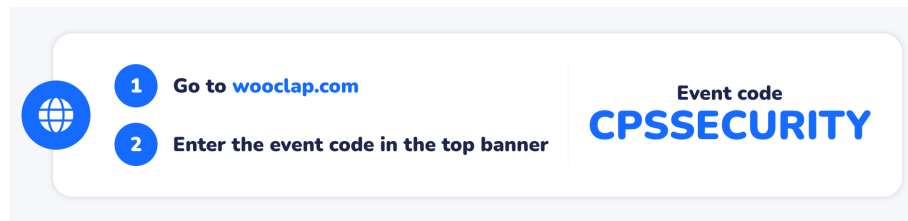


Course Evaluation

- **2-hour Written Examination (closed book)** **50%**
- **(New) Tutorial Attendance through Wooclap** **5%**
- **Quiz 1** **10%**
- **Quiz 2** **10%**
- **Lab 1+2 Report** **5%**
- **Lab 3+4 Report** **5%**
- **Project Report** **15%**
 - Project problems will be introduced during the labs
- Note 1: All practical sessions will be held at HW Lab 1
- Note 2: All practical and projects can be performed in a group of size 1/2/3
- Tutorial attendance marks
 - 5 marks: participation $\geq 70\%$ of tutorial classes
 - 2 marks: participation $\geq 50\%$ but $< 70\%$
 - 0 marks: participation $< 50\%$

Course Feedback, Discussions

- Your opinion matters
 - Provide feedback to the tutor/assistants on things you (dis)liked
 - Share your opinion through NTU student feedback system
- Discussion
 - It is important that you participate in course discussions
 - In classroom (through [wooclap](#) or after the lecture)
 - By Email/Course Discussion Board
 - By appointment with tutors/assistants



Attendance

Course Contents

- Introduction
 - Cyber Physical Systems, Internet-of-Things; Security
- Attack Surfaces
 - Network, Control, Computing, Storage
- Device-level Security
 - Microprocessors
- Secure Key Management
 - Distributed CPS, Biometrics
- Secure Communication
 - Communication protocols and attacks
- CPS Security Applications
 - Smart Card, Smart Grid, Smart Vehicle
- Guest Lectures (will not be tested)
 - Side-channel Attacks; AI Security; Post-Quantum Security

Reference Materials

- Books

- Handbook of Applied Cryptography by Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone, CRC press, fifth printing, August 2001, <http://cacr.uwaterloo.ca/hac/>
- Digital Integrated Circuits (2nd Edition), Jan M. Rabaey, A. Chandrakasan and B. Nikolic, Pearson, <https://www.amazon.com/Digital-Integrated-Circuits-2nd-Rabaey/dp/0130909963>
- Edward A. Lee and Sanjit A. Seshia, Introduction to Embedded Systems, A Cyber-Physical Systems Approach, Second Edition, <http://LeeSeshia.org>, ISBN 978-1-312-42740-2, 2015
- High-Performance Embedded Computing: Applications in Cyber-Physical Systems and Mobile Computing by Marilyn Wolf, Morgan Kaufmann, 2nd Edition, 2014

- Reference materials and links will be indicated for each topic in the slides/lecture notes.

Contents



- Course Organization

Cyber Physical Systems

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Where are the Cyber Physical Systems?



Principle I: Hidden

- Embedded Systems are designed to be “embedded”, i.e., these are not explicit like a desktop



- Example: for a sailing boat, embedded system is needed for
 - Wind direction/speed assessment/prediction
 - Current sailboat position/direction/speed/acceleration assessment
 - Competitors' position/speed assessment
 - Strategy evaluation and execution (speed/direction)

Tae Kwon Do: Who is the Winner?



The Electronic Scoring System



Where is the CPS?

Principle II: Everywhere

- Embedded Systems can be found/everywhere. It is pervasive.
- For a taekwondo armor, it is used for
 - Impact detection (scoring)
 - Concussion detection (medical emergency)
 - Heart rate, body vital sign detection, audio/video recording (training)

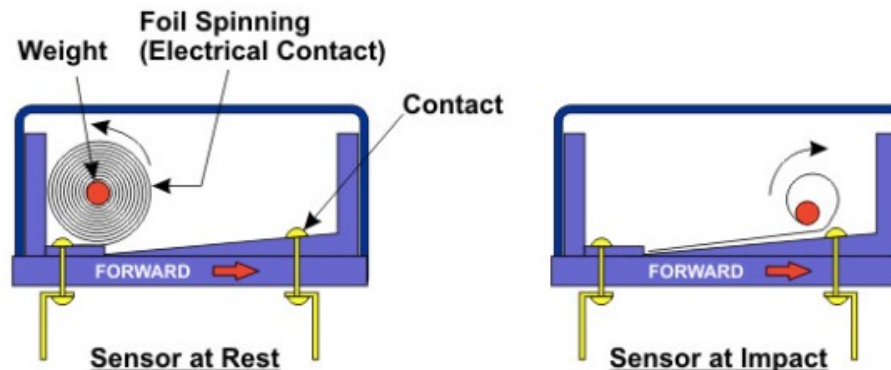
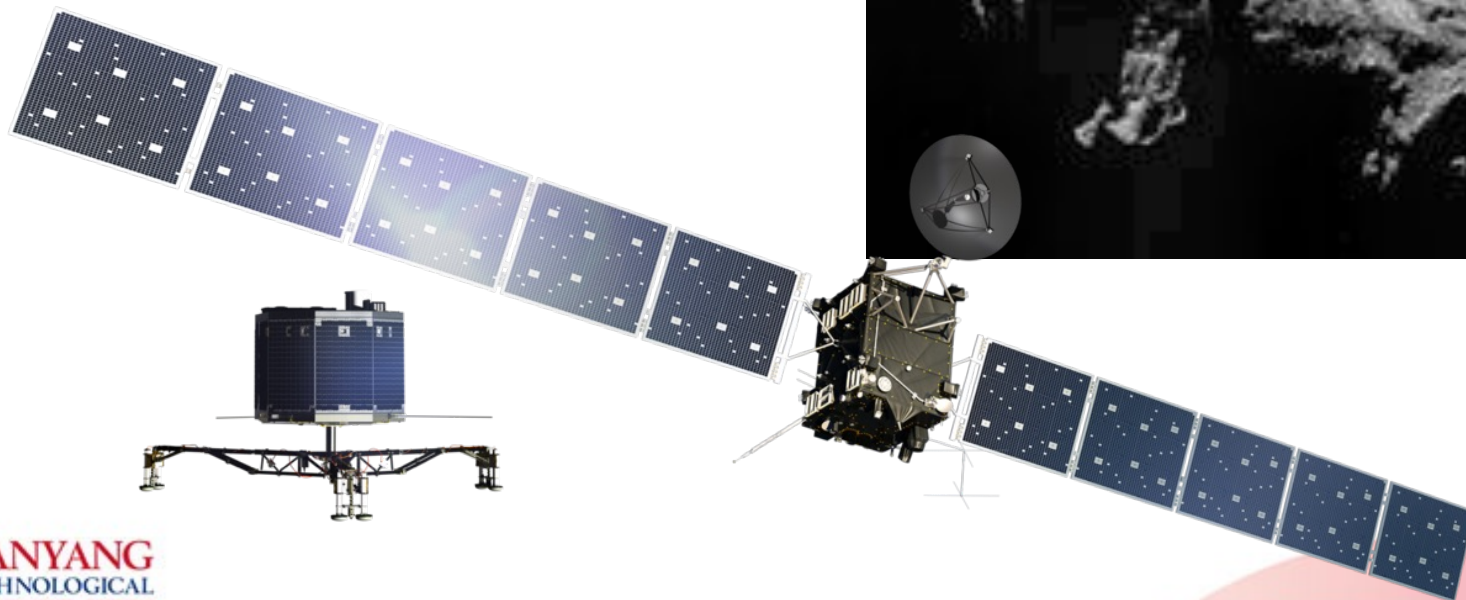
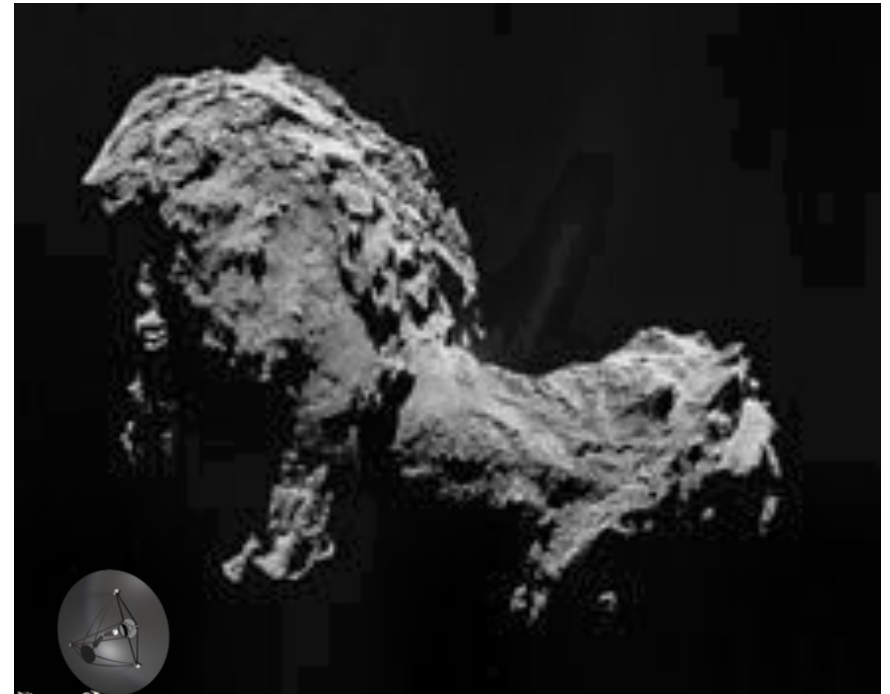


Figure 2. Functional principle to a typical roller type airbag sensor. Source: Erjavec, J. (2010). *Automotive Technology: A Systems Approach*. New York: Delmar, Cengage Learning.

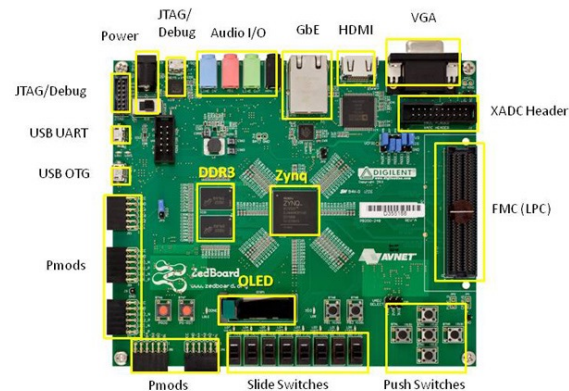
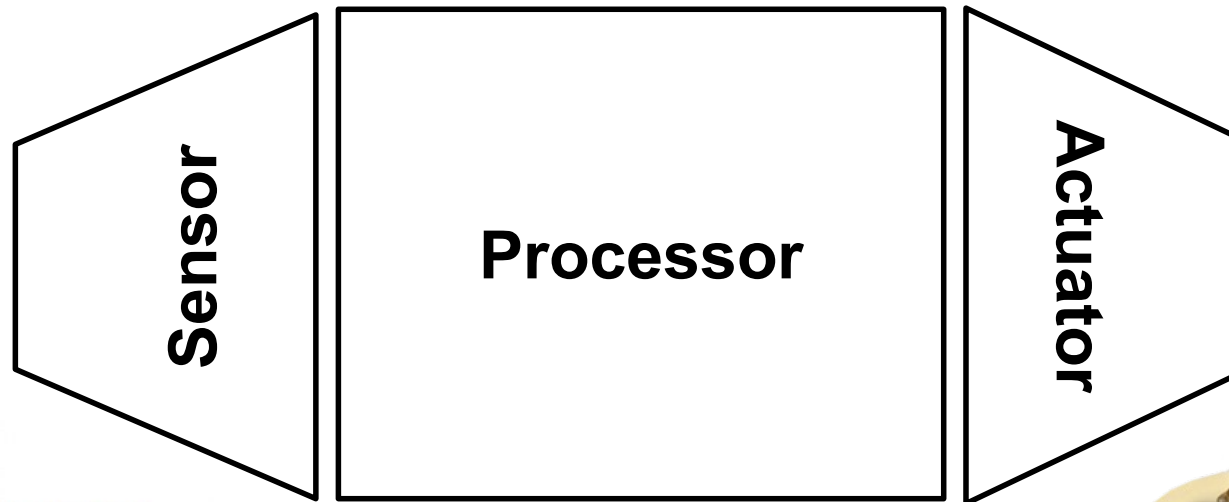
Principle III: Complex

- Comet Churyumov-Gerasimenko
- Satellite Rosetta (Launched 2004)
- Lander Philae (Landed 2014)



Embedded Systems

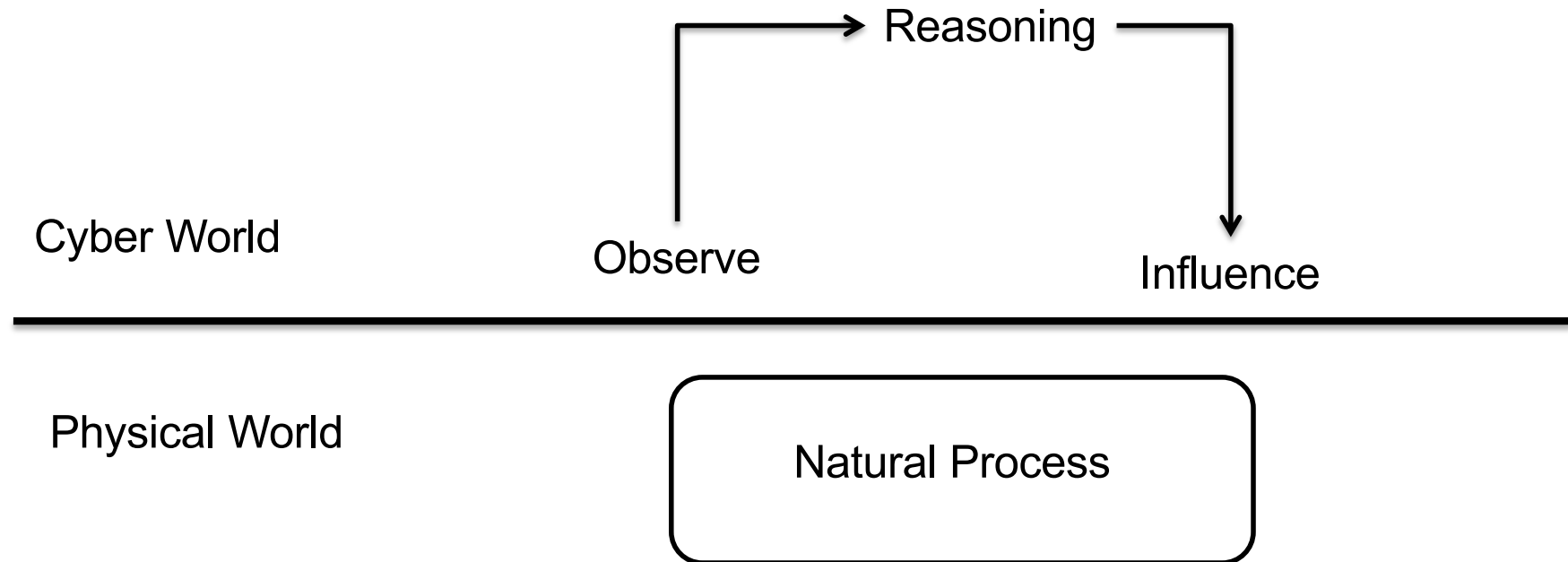
- **Complex, Hidden, Everywhere**



* SD card cage and QSPI Flash reside on backside of board



Why Cyber-Physical Systems



Characteristics of CPS

- Must be **robust**:
- **Reliability**: $R(t)$ = probability of system working correctly, provided that it was working at $t=0$
- **Maintainability**: $M(d)$ = probability of system working correctly d time units after error occurred.
- **Availability**: probability of system working at time t
- **Safety**: no harm to be caused
- **Security**: confidential and authenticated communication, control and storage

Characteristics of CPS (*contd.*)

- ***Must be efficient:***
 - Energy efficient
 - Code-size and data memory efficient
 - Run-time efficient
 - Weight efficient
 - Cost efficient
- ***Dedicated towards a certain application:*** Knowledge about behavior at design time can be used to minimize resources and to maximize robustness.



Characteristics of CPS (*contd.*)

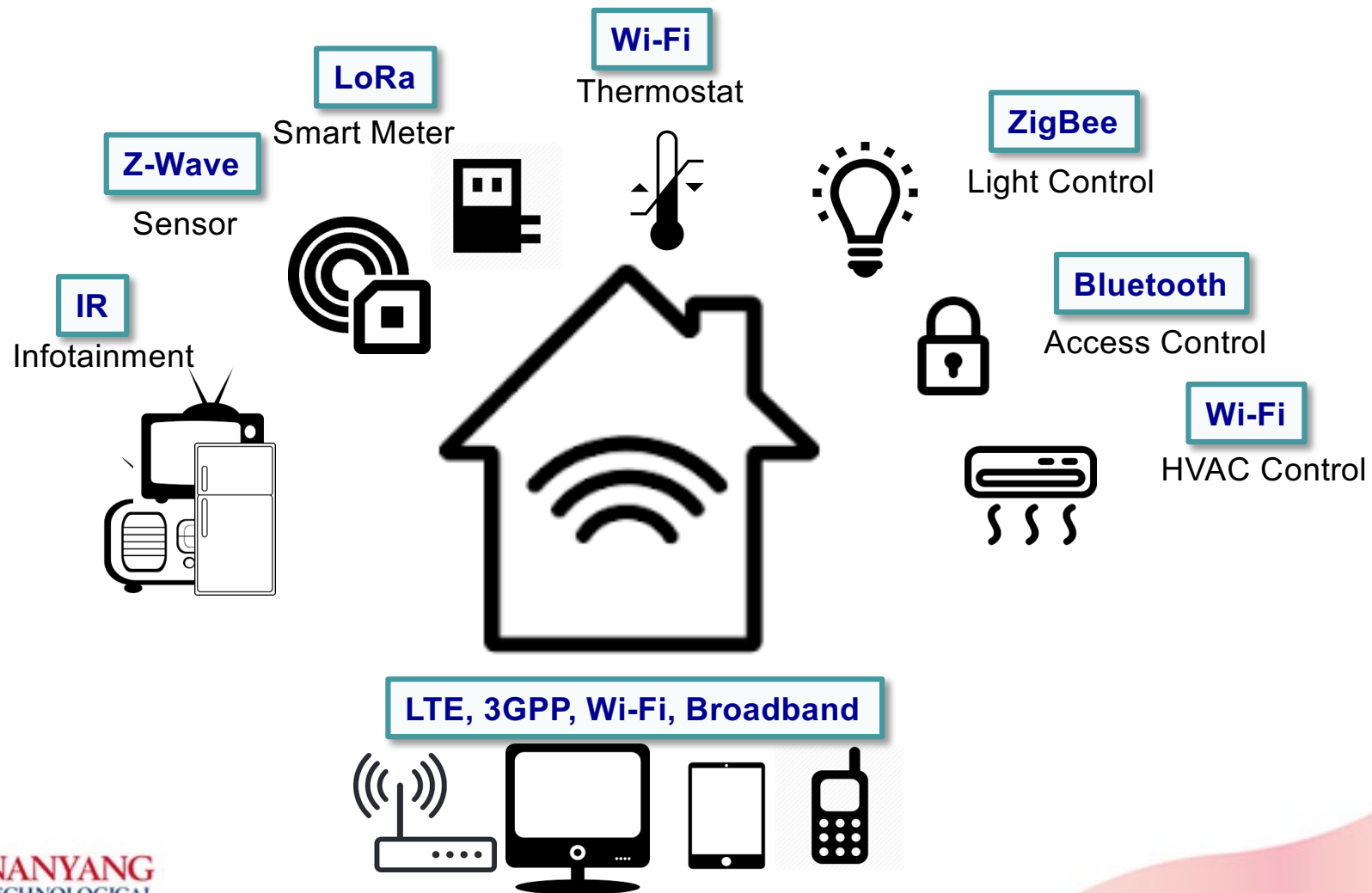
- Many CPS *must meet real-time constraints*:
 - A real-time system must react to stimuli from the controlled object (or the operator) within the time interval dictated by the environment.
- For real-time systems, right answers arriving too late are wrong.
 - "A real-time constraint is called **hard**, if not meeting that constraint could result in a catastrophe" [Kopetz, 1997].
- All other time-constraints are called **soft**.



What about Internet-of-Things (IoT) ?

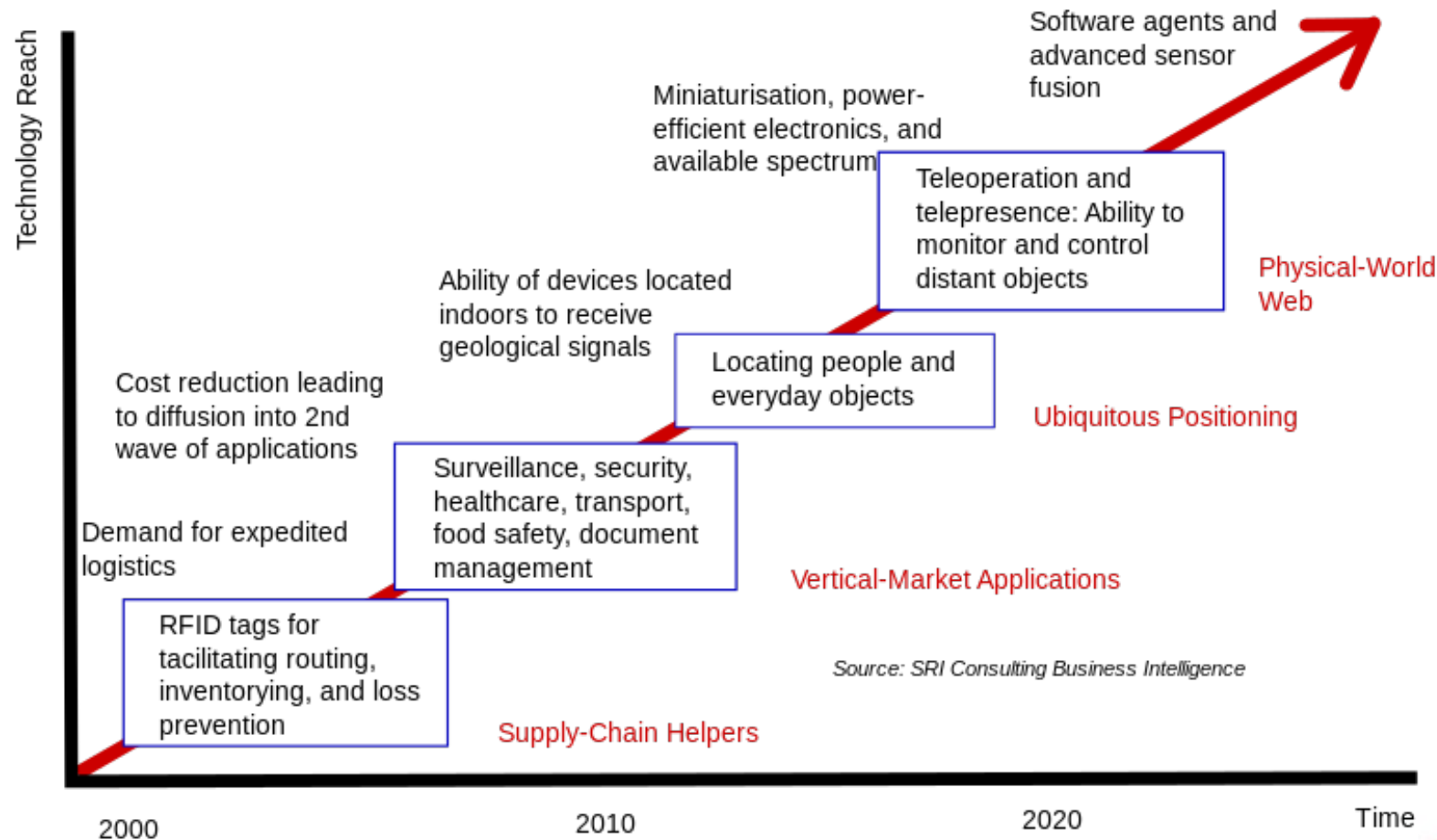


Internet of Things (IoT)



IoT Proliferation

Technology roadmap: The Internet of Things



IoT + AI: Distributed and Autonomous Agents

- Spoofing cameras through adversarial objects
 - Multiple cameras is a potential solution but, no guarantee
 - Explored hybrid solutions, including color palette reduction



Cao, Y., et al *Invisible for both Camera and LiDAR: Security of Multi-Sensor Fusion based Perception in Autonomous Driving Under Physical-World Attacks*. 2021 IEEE Symposium on Security and Privacy (S&P)

What did we learn so far?

- What is Cyber-Physical Systems/Internet-of-Things ?
- Where can we find it?
- How to design/secure one?
 - ***Next slides...***



Contents



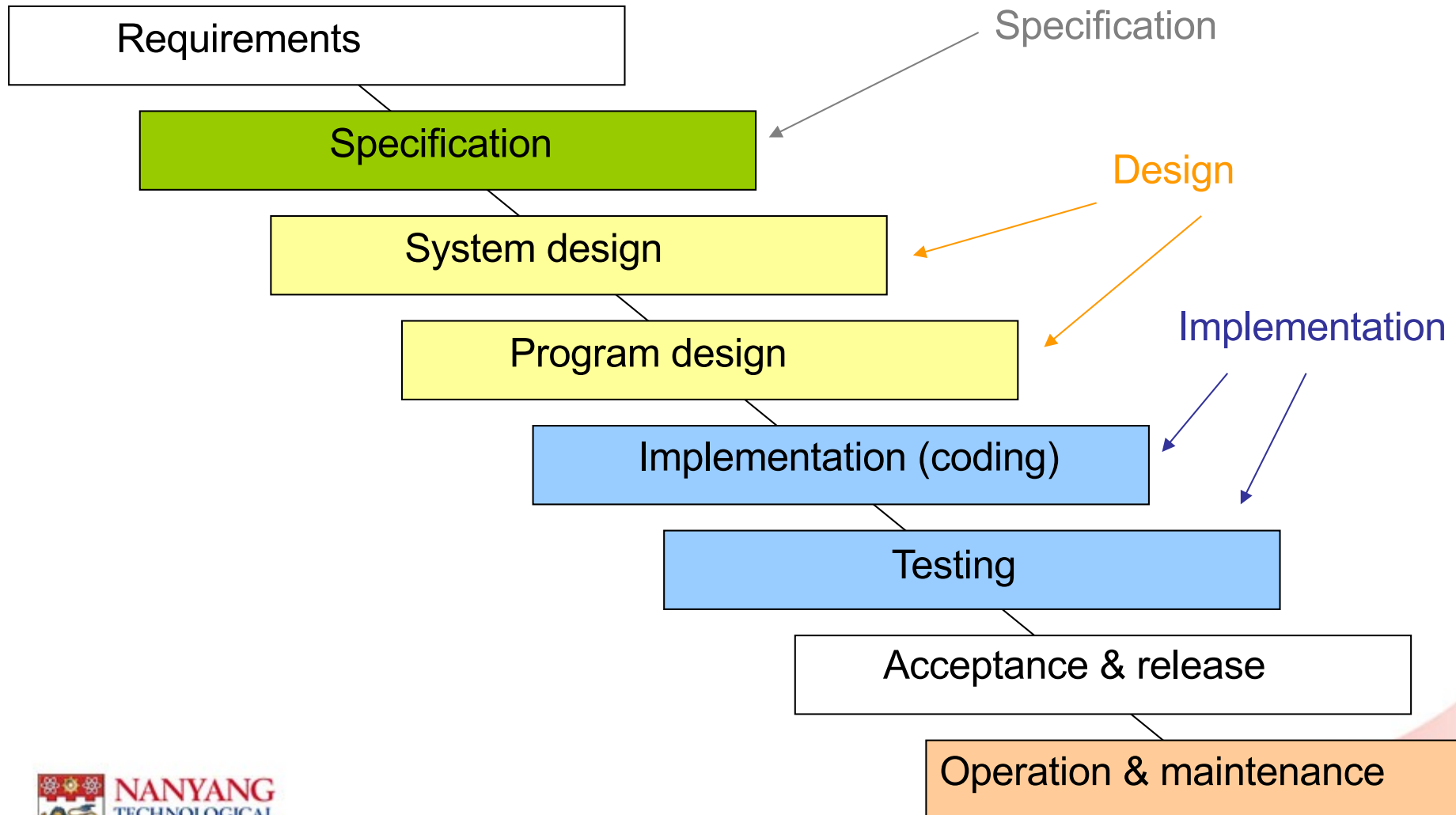
- Course Organization

Cyber Physical Systems

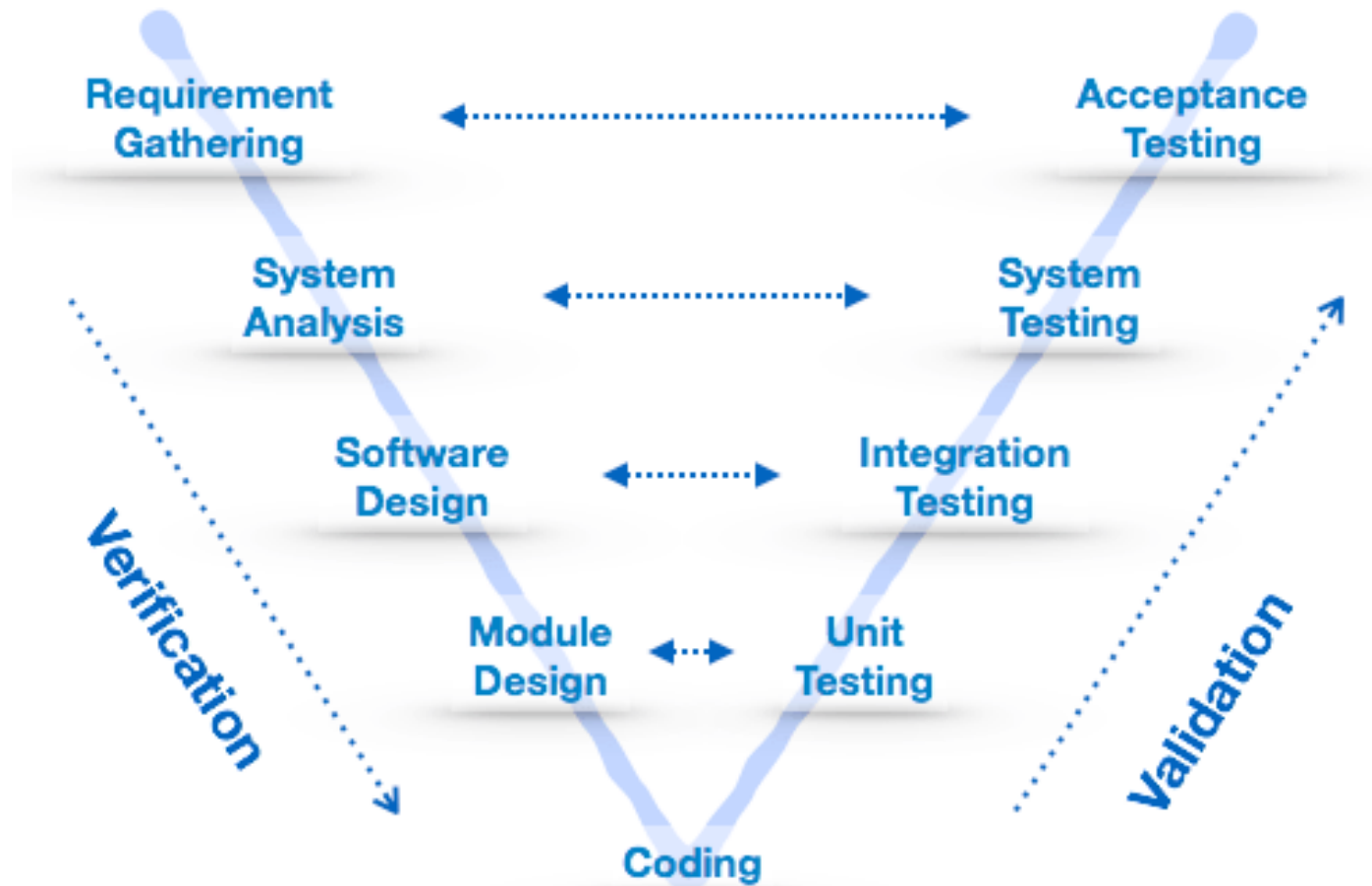
- ***Design Techniques***
- Security
- Discussion



Design Technique: Waterfall Model

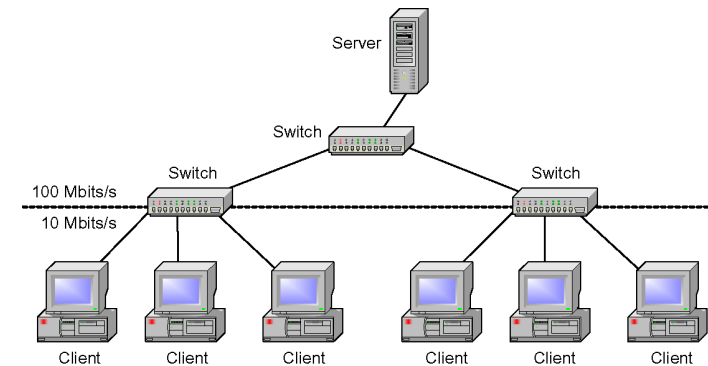
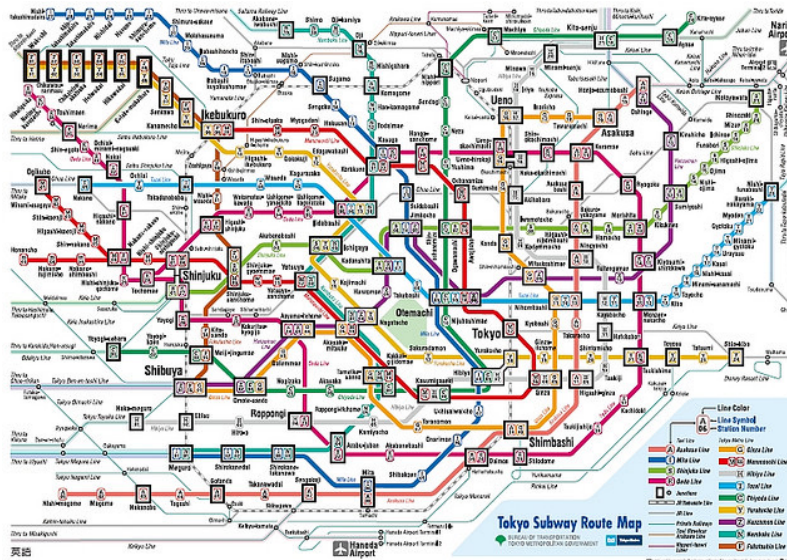


Design Technique: V Model



Specification

- Represent hierarchical computation and communication
 - Humans are not capable of understanding systems containing more than a few objects
 - Most actual systems require more objects → Hierarchy

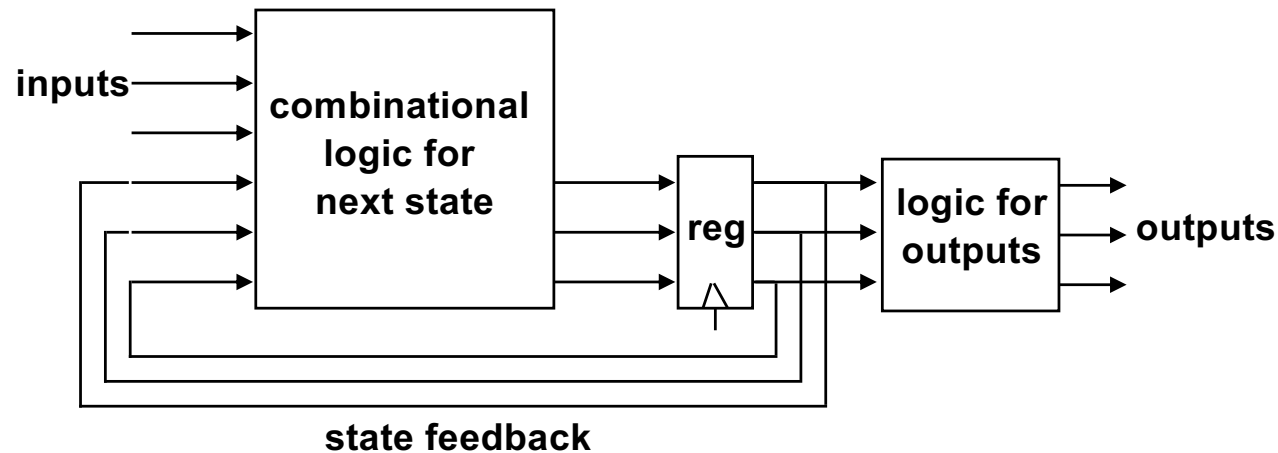


FSMs: Mealy/Moore

- Moore/Mealy machines
- These are two different ways to express the FSMs with respect to the output.
- Both have different advantages so it is good to know them.



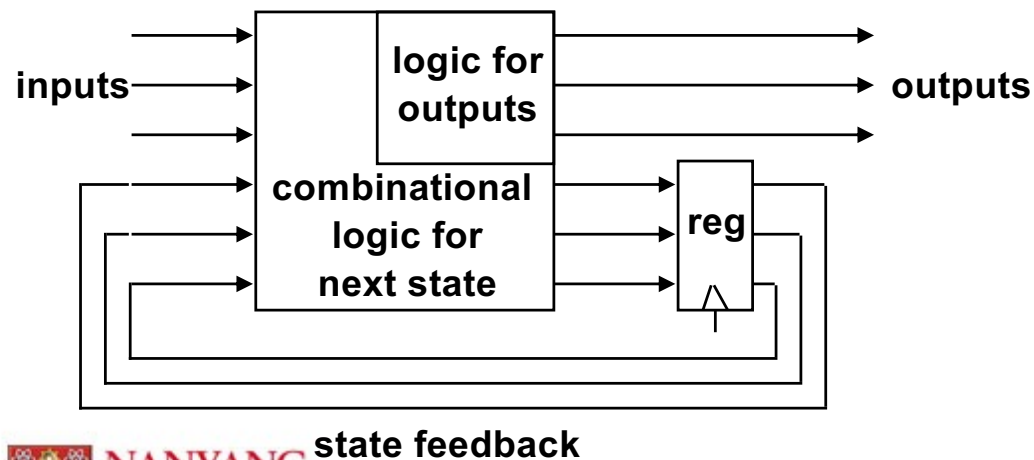
Moore versus Mealy machines



Moore machine

Outputs are a function of current state

Outputs change synchronously with state changes



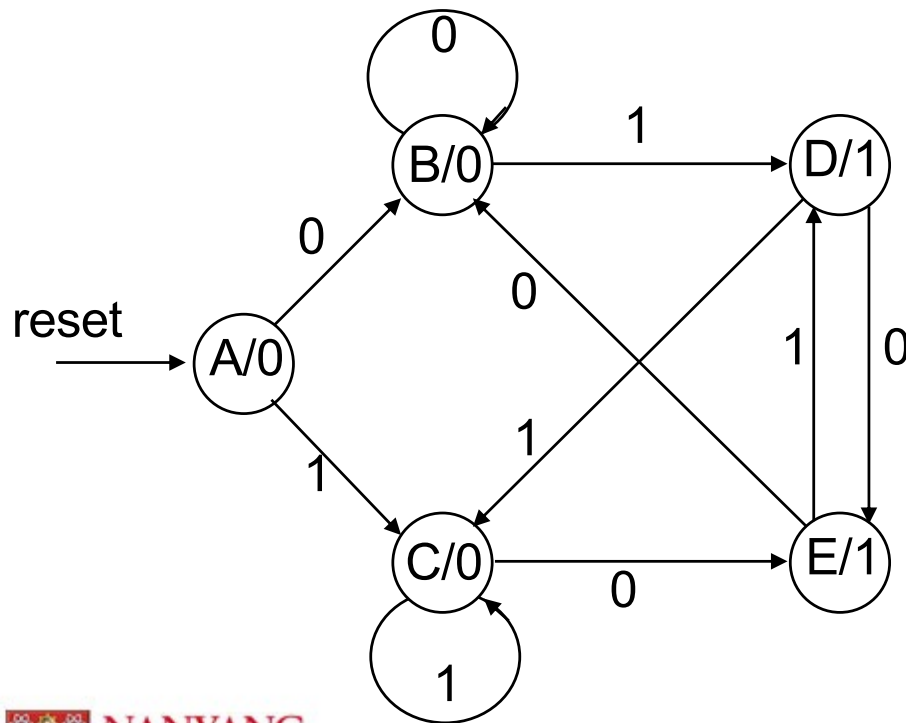
Mealy machine

Outputs depend on state **and** on inputs

Input changes can cause immediate output changes (**asynchronous**)

Example “01 or 10” detector: a Moore machine

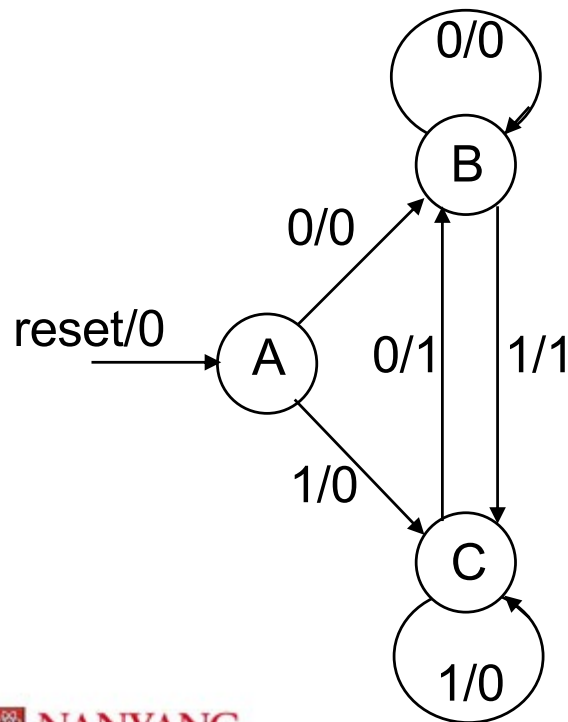
- Output is a function of state only
 - Specify output in the state bubble



reset	input	current state	next state	current output
1	—	—	A	0
0	0	A	B	0
0	1	A	C	0
0	0	B	B	0
0	1	B	D	0
0	0	C	E	0
0	1	C	C	0
0	0	D	E	1
0	1	D	C	1
0	0	E	B	1
0	1	E	D	1

Example “01 or 10” detector: a Mealy machine

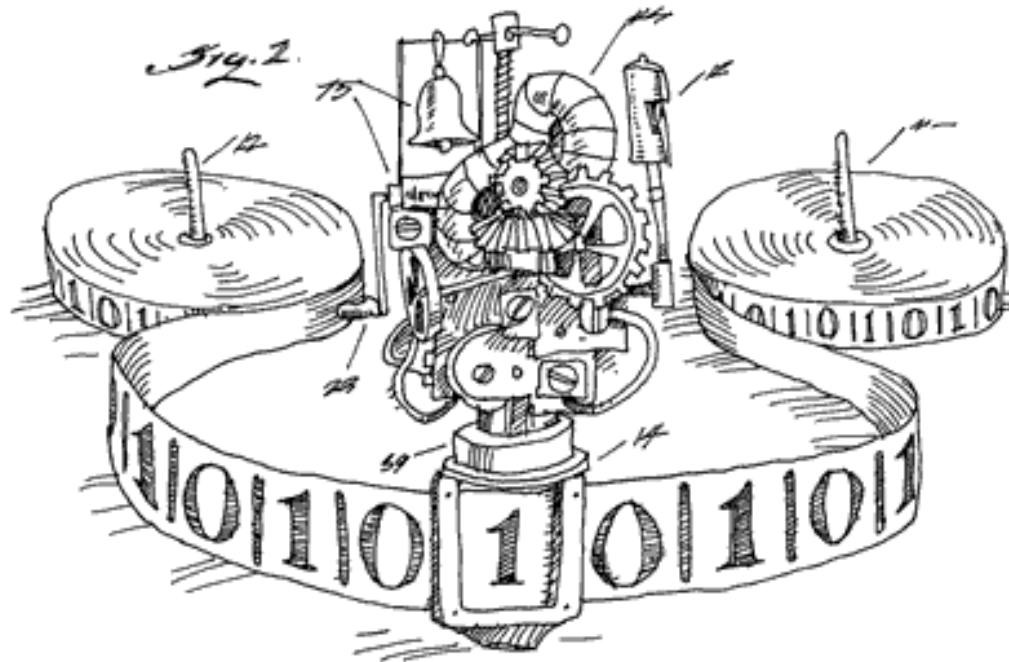
- Output is a function of state and inputs
 - Specify outputs on transition arcs



reset	input	current state	next state	current output
1	—	—	A	0
0	0	A	B	0
0	1	A	C	0
0	0	B	B	0
0	1	B	C	1
0	0	C	B	1
0	1	C	C	0

Component

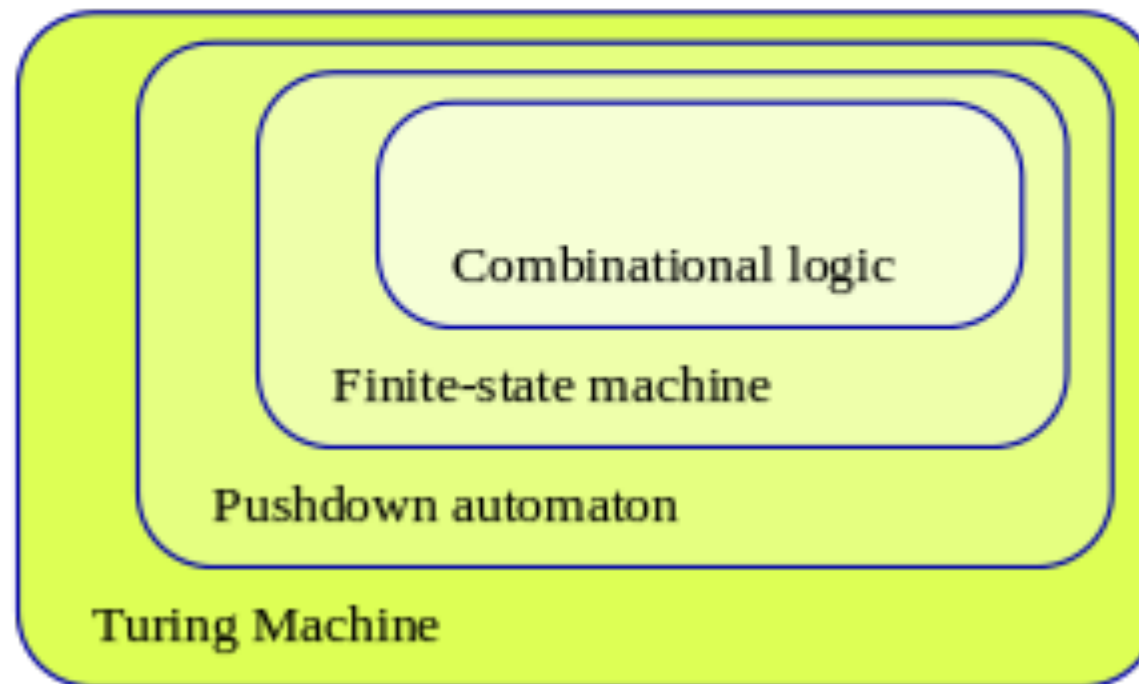
- Turing Machine



On Computable Numbers, With An Application To The Entscheidungsproblem, by A. M. Turing, 1936

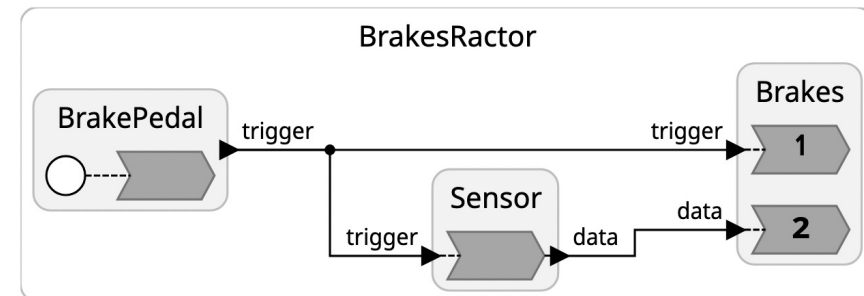
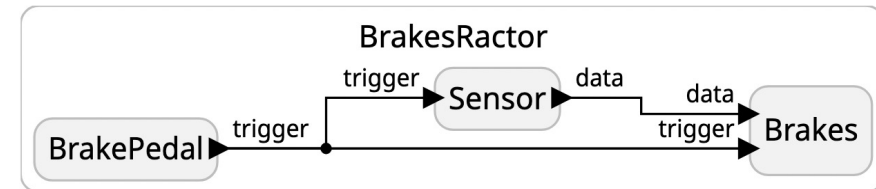
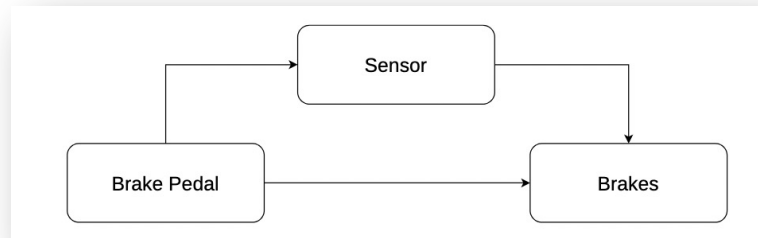
Computing power of components

Automata theory



Model-based Design

- Described using System-level Design Languages, e.g., Lingua Franca (<https://www.lf-lang.org/>)



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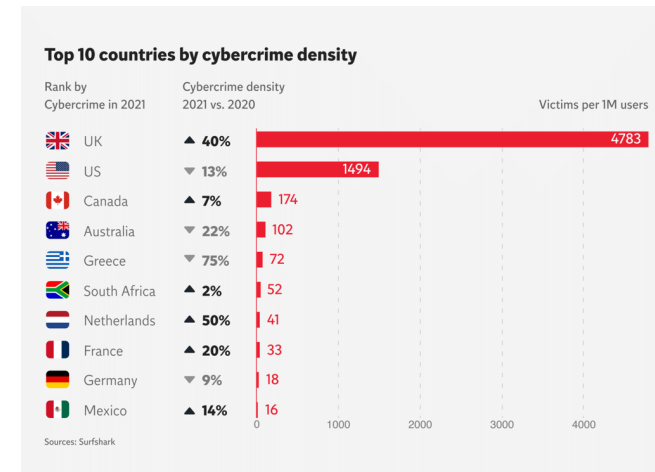
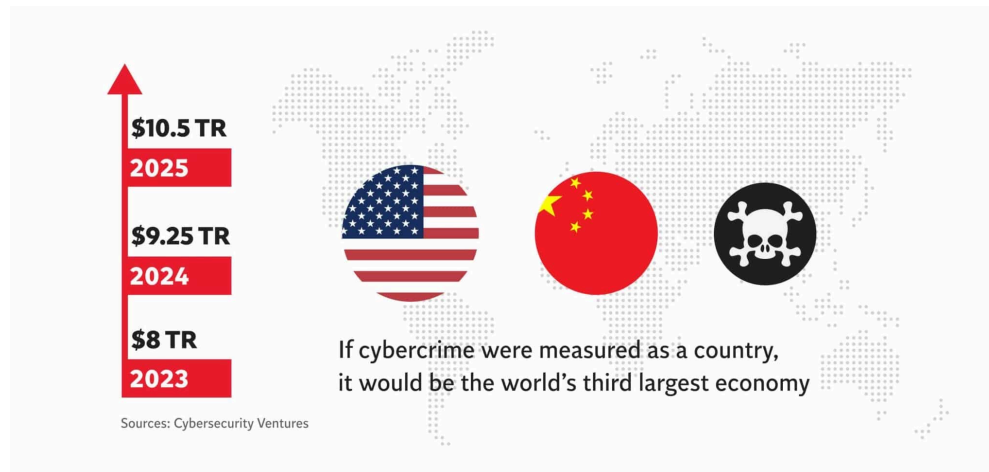
Security

- Discussion



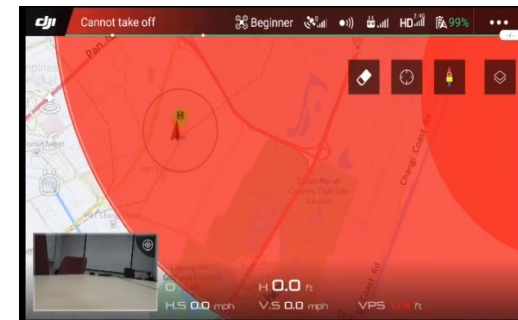
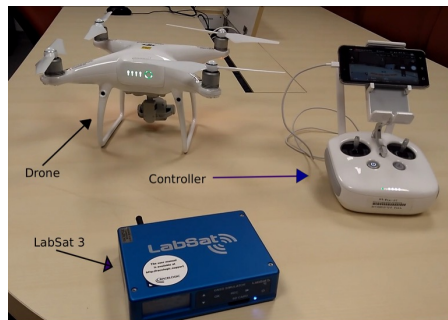
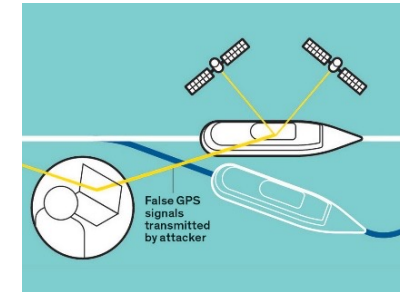
Cyber Crime

- Global annual cost of cyber crime to reach **\$8 Trillion/year** in 2023.



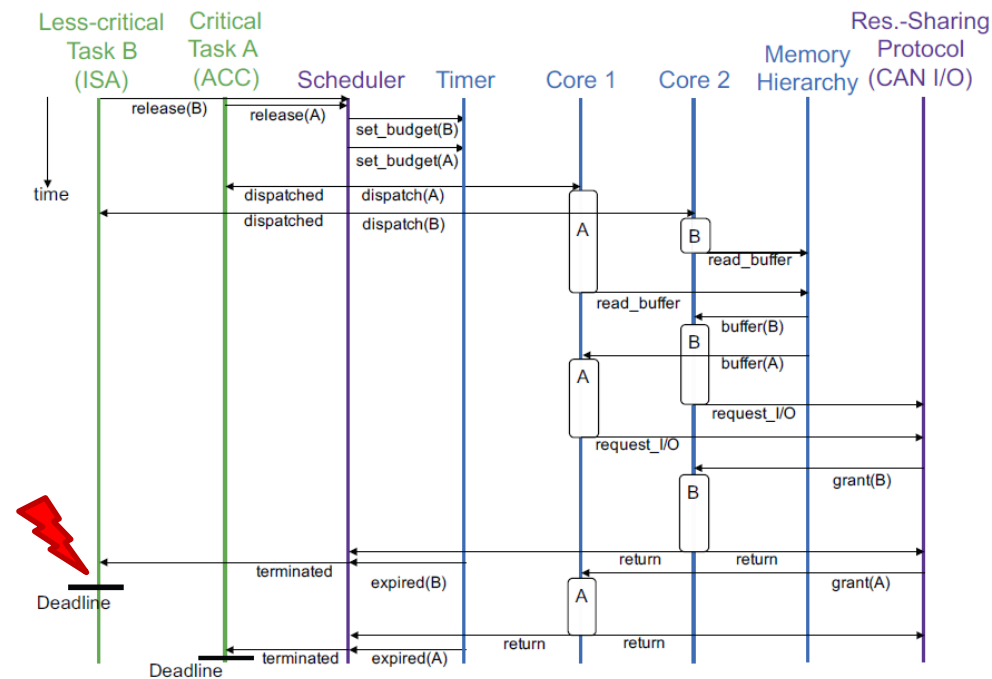
CPS/IoT Systems View

- Example: GPS Spoofing



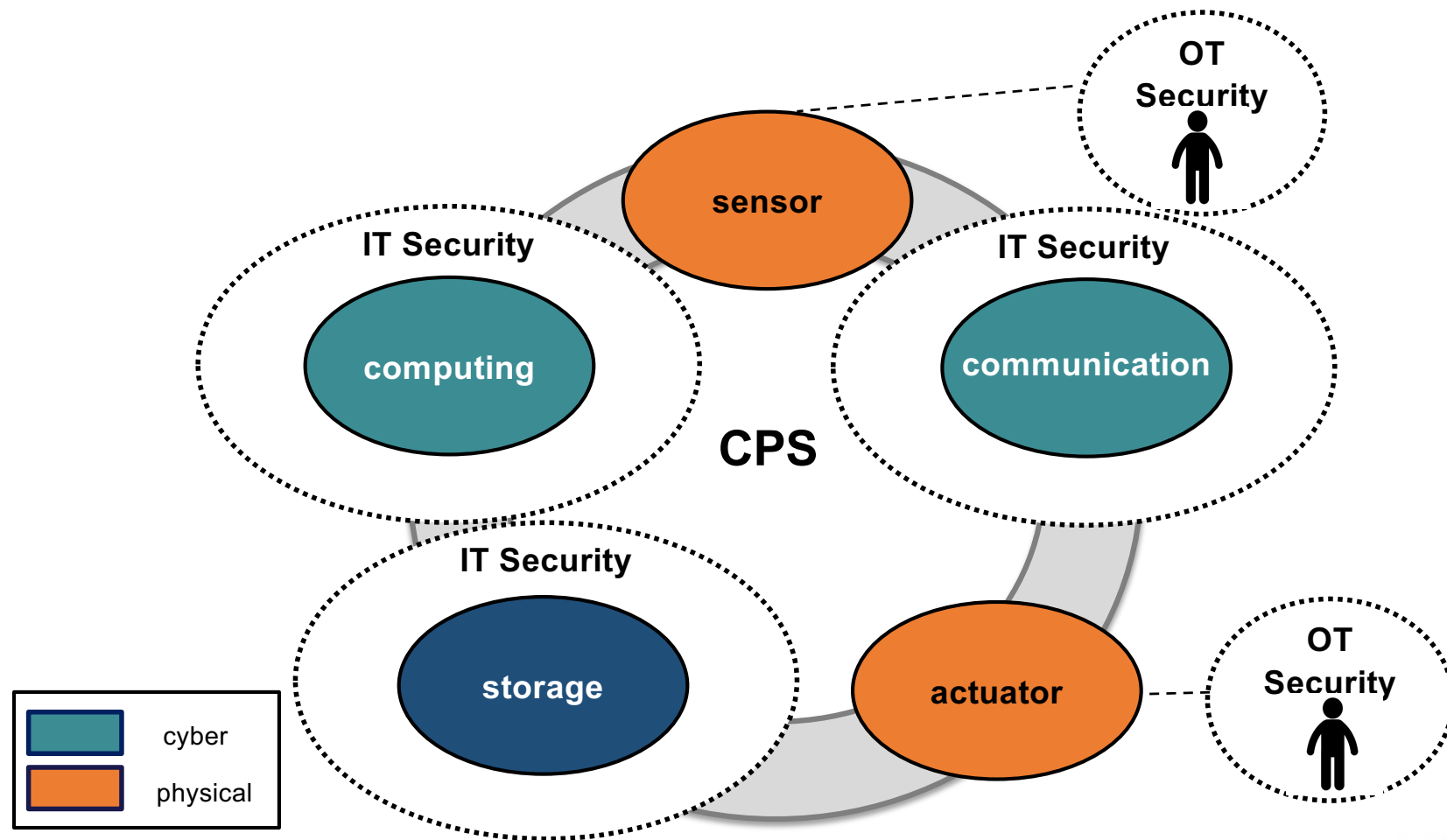
Real-Time Systems View

- Example: Deadline Miss



Arvind Easwaran, Anupam Chattopadhyay and Shivam Bhasin: A systematic security analysis of real-time cyber-physical systems. [ASP-DAC 2017](#)

CPS Security View



What is Privacy?

- Definitions

- “*Privacy is a basic psychological need*” S. M. Jourard, 1966
- “*Privacy is necessary for managing social ties*”, Chaikin, 1977
- “*Privacy is critical for flexibly projecting and designing our self to others*”, Goffman 1959, Vitak 2015

- Privacy Laws



France fines Google and Facebook
€210m over user tracking

Data privacy watchdog says websites make it difficult for users to
refuse cookies



China fines Didi \$1.2 billion for violating
cybersecurity and data laws

By Yong Xiong, Larry Register and Laura Ho, CNN Business
© 3 minute read · Updated 9:07 AM EDT, Thu July 21, 2022



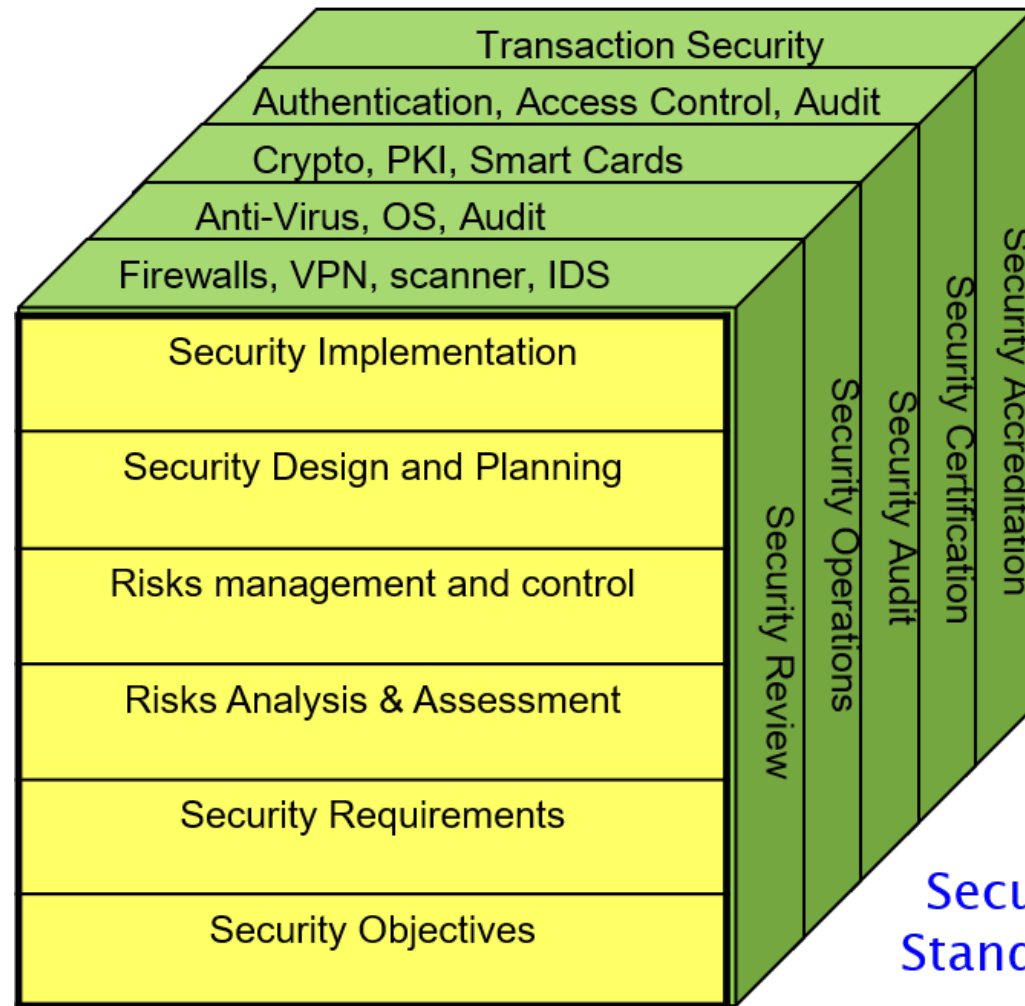
TVR Analysis

Rating system to quantify the security hazards

- **Threat**
 - External/Internal; Human/Machine
 - Example: Inadequately trained staff, Overheating
- **Vulnerability**
 - Exposure: Affects one/multiple components
 - Severity: Requires little/heavy resources to exploit, huge/minor loss. Cost/Impact Ratio and Adversarial Model.
- **Risk**
 - Organizational Impact. Economic, Brand Value, Life-threatening

Security Management

Security Technology



Security
Standards

What did we learn?

- **What is Model of Computation?**
 - What is CPS?
 - Components (FSM, Turing)
 - *Is your computer a Turing machine?*
- **What are the real-world implications of security?**
 - CPS security perspective from IT and OT
 - Real-time system perspective
 - TVR Model
 - Privacy



Further Reading

- Turing Machines, Automata Theory
 - http://www.doc.ic.ac.uk/~imh/teaching/Turing_machines/240.pdf
 - https://en.wikipedia.org/wiki/Turing_machine
 - <http://theory.stanford.edu/~trevisan/cs154-12/turing-machines-1.pdf>
 - Follow how a multi-tape Turing machine is proved to be equivalent to a Turing machine
- SANS Institute: Threat and Risk Assessment
 - <https://www.sans.org/reading-room/whitepapers/auditing/overview-threat-risk-assessment-76>



The End

